

NUCLEAR PHYSICS MCQ

1. Nuclear physics primarily studies structure formation stability of

- a) electron clouds
- b) atomic nuclei
- c) chemical bonds
- d) molecular orbits

Correct answer: b

2. Radioactivity discovered 1896 by Henri Becquerel during studies of

- a) fluorescence
- b) uranium phosphorescence
- c) radium compounds
- d) neutron effects

Correct answer: b

3. Pierre Marie Curie discovered radioactive elements

- a) uranium thorium
- b) polonium radium
- c) plutonium neptunium
- d) actinium protactinium

Correct answer: b

4. Radioactivity defined spontaneous disintegration unstable

- a) electron shells
- b) isotope nucleus
- c) molecular bonds
- d) proton clusters

Correct answer: b

5. Radioactive elements unstable due excess

- a) electrons
- b) nuclear charge
- c) neutrons only
- d) protons only

Correct answer: b

6. Common natural radioactive elements include all except

- a) uranium
- b) radium
- c) polonium
- d) carbon-12

Correct answer: d

7. Natural radioactivity spontaneous decay elements atomic number above

- a) 82
- b) 92
- c) 50
- d) 100

Correct answer: a

8. Artificial radioactivity induced bombarding stable nuclei with

- a) alpha particles
- b) neutrons
- c) gamma rays
- d) positrons

Correct answer: b

9. Artificial radioactive elements atomic numbers range

- a) 1-82

b) 83-94

c) 95-118

d) above 119

Correct answer: c

10. Parent nucleus decays producing

a) identical nucleus

b) daughter nucleus

c) proton shower

d) electron cascade

Correct answer: b

11. Curie traditional radioactivity unit equals 3.7×10^{10}

a) decays minute

b) disintegrations second

c) fissions hour

d) reactions day

Correct answer: b

12. Becquerel SI unit defined one

a) curie

b) rutherford

c) disintegration second

d) roentgen kg

Correct answer: c

13. Roentgen measures radiation ionizing

a) water

b) metals

c) air volume

d) solids

Correct answer: c

14. Alpha radiation consists helium

a) atoms

b) nuclei

c) electrons

d) photons

Correct answer: b

15. Beta radiation high energy

a) protons

b) neutrons

c) electrons

d) positrons only

Correct answer: c

16. Gamma radiation electromagnetic

a) particles

b) waves

c) nuclei

d) electrons

Correct answer: b

17. Alpha rays highest

a) penetrating power

b) speed

c) ionizing power

d) range vacuum

Correct answer: c

18. Radioactive displacement law alpha decay mass decreases by

- a) 2
- b) 4
- c) 0
- d) 1

Correct answer: b

19. Beta decay atomic number

- a) decreases 2
- b) increases 1
- c) unchanged
- d) halves

Correct answer: b

20. Gamma decay affects only nuclear

- a) mass
- b) charge
- c) energy level
- d) size

Correct answer: c

21. Nuclear fission heavy nucleus splits lighter

- a) electrons
- b) nuclei
- c) protons
- d) neutrons

Correct answer: b

22. Primary fissionable isotope uranium

- a) U-238

b) U-235

c) U-239

d) U-240

Correct answer: b

23. U-238 non-fissionable called

a) fertile

b) stable

c) reactive

d) explosive

Correct answer: a

24. Fertile materials convert fissionable include

a) U-235

b) Th-232

c) Pu-239

d) Rn-222

Correct answer: b

25. Chain reaction neutron causes fission releasing more

a) protons

b) neutrons

c) electrons

d) photons

Correct answer: b

26. Controlled chain reaction maintains steady

a) explosion

b) fission rate

c) fusion

d) decay

Correct answer: b

27. Critical mass minimum sustain

a) fusion

b) chain reaction

c) absorption

d) moderation

Correct answer: b

28. Atom bomb supercritical mass produces uncontrolled

a) fusion

b) fission explosion

c) decay

d) capture

Correct answer: b

29. Hydrogen bomb triggers fusion using built-in

a) reactor

b) atom bomb

c) laser

d) magnet

Correct answer: b

30. Mass defect difference actual calculated

a) volume

b) nucleon masses

c) charge

d) energy levels

Correct answer: b

31. Fusion requires temperatures million

- a) 10^3
- b) 10^7
- c) 10^5
- d) 10^9

Correct answer: b

32. Stellar energy originates nuclear

- a) fission
- b) fusion
- c) fission-fusion
- d) decay

Correct answer: b

33. Radio-iodine I-131 treats

- a) cancer
- b) goitre
- c) anaemia
- d) heart disease

Correct answer: b

34. Co-60 used cancer

- a) diagnosis
- b) therapy
- c) tracing
- d) sterilization

Correct answer: b

35. P-32 agriculture increases crop

- a) colour

b) productivity

c) size

d) taste

Correct answer: b

36. Gamma radiography detects metal

a) thickness

b) cracks flaws

c) density

d) colour

Correct answer: b

37. Carbon-14 dating estimates age

a) metals

b) organic remains

c) rocks

d) stars

Correct answer: b

38. Safe annual radiation limit 20

a) roentgen

b) millisievert

c) curie

d) becquerel

Correct answer: b

39. 600 R exposure likely causes

a) nausea

b) death

c) cancer only

d) no harm

Correct answer: b

40. Nuclear reactor heat power

a) fusion plant

b) fission plant

c) chemical plant

d) solar plant

Correct answer: b

41. Moderator slows fast

a) protons

b) neutrons

c) electrons

d) gamma rays

Correct answer: b

42. Control rods material absorbs

a) heat

b) neutrons

c) protons

d) heat coolant

Correct answer: b

43. Coolant transfers heat produce

a) light

b) steam electricity

c) chemicals

d) magnets

Correct answer: b

44.First Indian reactor

- a) Dhruva
- b) Apsara
- c) Cirus
- d) Purnima

Correct answer: b

45.India nuclear reactors operating

- a) 10
- b) 22
- c) 30
- d) 50

Correct answer: b

46.PHWRs India number

- a) 4
- b) 18
- c) 22
- d) 8

Correct answer: b

47.Kalpakkam Tamil Nadu uses

- a) LWR
- b) PHWR
- c) breeder
- d) BWR

Correct answer: b

48.Kudankulam reactors

- a) PHWR

- b) LWR
- c) graphite
- d) gas-cooled

Correct answer: b

49. BARC named after

- a) Vikram Sarabhai
- b) Homi Bhabha
- c) CV Raman
- d) APJ Abdul Kalam

Correct answer: b

50. India's nuclear capacity approx

- a) 2000 MWe
- b) 6780 MWe
- c) 10000 MWe
- d) 5000 MWe

Correct answer: b

51. Natural radioactivity major elements earth's crust

- a) carbon oxygen
- b) uranium thorium
- c) iron silicon
- d) sodium calcium

Correct answer: b

52. Artificial radioactivity f-block elements

- a) 1-57
- b) 58-71
- c) 95-118

d) 89-103

Correct answer: c

53. Rutherford unit 10^6 disintegrations

a) minute

b) second

c) hour

d) day

Correct answer: b

54. Alpha rays deflected magnetic field

a) least

b) most

c) same beta

d) undeflected

Correct answer: b

55. Penetrating power order

a) $\alpha > \beta > \gamma$

b) $\alpha < \beta < \gamma$

c) $\beta > \alpha > \gamma$

d) $\gamma > \alpha > \beta$

Correct answer: b

56. Displacement law gamma emission changes

a) A Z both

b) A Z unchanged

c) A only

d) Z only

Correct answer: b

57.U-235 fission triggered absorbing

- a) proton
- b) neutron
- c) alpha
- d) gamma

Correct answer: b

58.Subcritical mass chain reaction

- a) explodes
- b) sustains
- c) dies out
- d) fuses

Correct answer: c

59.Supercritical mass leads

- a) steady power
- b) rapid explosion
- c) slow decay
- d) absorption

Correct answer: b

60.Thermonuclear fusion needs high

- a) pressure
- b) temperature
- c) density confinement
- d) all above

Correct answer: d

61. Hydrogen bomb energy release fission bomb by

- a) 10 times
- b) 1000 times
- c) same
- d) less

Correct answer: b

62. Na-24 traces

- a) thyroid
- b) blood circulation
- c) bones
- d) tumours

Correct answer: b

63. Fe-59 diagnoses

- a) cancer
- b) anaemia
- c) heart
- d) goitre

Correct answer: b

64. Californium-252 detects airport

- a) metals
- b) explosives
- c) drugs
- d) liquids

Correct answer: b

65. Americium-241 smoke

- a) tracers

- b) detectors
- c) therapy
- d) sterilization

Correct answer: b

66. Radiation safety three principles

- a) time distance shielding
- b) heat light sound
- c) mass charge energy
- d) speed force motion

Correct answer: a

67. Protection wall nuclear reactor concrete

- a) steel
- b) lead lined
- c) glass
- d) wood

Correct answer: b

68. Breeder reactor produces more than consumes

- a) waste
- b) fissile material
- c) heat
- d) neutrons

Correct answer: b

69. First nuclear reactor built

- a) 1939 Germany
- b) 1942 Chicago
- c) 1945 Japan

d) 1950 India

Correct answer: b

70. India's nuclear power ranking world

a) first

b) fifth

c) tenth

d) twentieth

Correct answer: b

71. Atomic Energy Commission established

a) 1947

b) 1948

c) 1950

d) 1952

Correct answer: b

72. Fast breeder reactor converts U-238

a) stable

b) Pu-239

c) waste

d) thorium

Correct answer: b

73. Pressurized water reactor uses

a) heavy water

b) light water coolant

c) graphite moderator

d) air coolant

Correct answer: b

74. Boron rods absorb neutrons prevent

- a) overheating
- b) explosion
- c) meltdown
- d) all above

Correct answer: d

75. Heavy water moderator better than graphite because

- a) cheaper
- b) slows neutrons effectively
- c) produces heat
- d) radioactive

Correct answer: b

76. Radio-phosphorus P-32 treats

- a) cancer
- b) skin diseases
- c) anaemia
- d) heart

Correct answer: b

77. Krypton-85 high intensity

- a) bulbs
- b) discharge lamps
- c) tracers
- d) dating

Correct answer: b

78. Promethium-147 portable

- a) reactors

b) X-ray units

c) bombs

d) fuels

Correct answer: b

79.Safe weekly roentgen limit

a) 20 mR

b) 100 mR

c) 600 R

d) 1000 R

Correct answer: b

80.Acute Radiation Syndrome high dose causes

a) tanning

b) nausea vomiting

c) strength

d) sleepiness

Correct answer: b

81.Time minimization reduces radiation

a) source strength

b) exposure dose

c) shielding need

d) distance effect

Correct answer: b

82.Distance doubling reduces dose by

a) half

b) quarter

c) double

d) same

Correct answer: b

83. Gloves handling radioactive materials prevent

a) heat

b) contamination

c) light

d) sound

Correct answer: b

84. No eating drinking radioactive labs prevents

a) spills

b) internal contamination

c) heat

d) shocks

Correct answer: b

85. Cirus reactor India located

a) Kalpakkam

b) Trombay

c) Tarapur

d) Rajasthan

Correct answer: b

86. Dhruva research reactor uses

a) uranium

b) heavy water

c) both

d) light water

Correct answer: c

87. Purnima fast breeder

- a) research reactor
- b) power reactor
- c) weapon reactor
- d) medical reactor

Correct answer: a

88. India's nuclear states exclude

- a) Maharashtra
- b) Kerala
- c) Rajasthan
- d) Gujarat

Correct answer: b

89. NPCIL operates India's

- a) research reactors
- b) power reactors
- c) weapon reactors
- d) medical reactors

Correct answer: b

90. Radio-gold Au-198 cancer

- a) diagnosis
- b) skin treatment
- c) heart tracing
- d) bone scanning

Correct answer: b

91. Thickness gauging uses

- a) alpha

b) beta radiation

c) gamma

d) neutrons

Correct answer: b

92. Luminescent paints contain

a) radium

b) tritium phosphor

c) uranium

d) plutonium

Correct answer: b

93. Hiroshima Nagasaki bombs

a) hydrogen

b) fission uranium

c) fusion

d) neutron

Correct answer: b

94. Fusion energy binding per nucleon maximum

a) light elements

b) iron group

c) heavy elements

d) uranium

Correct answer: b

95. Natural series end stable

a) lead

b) uranium

c) thorium

d) radium

Correct answer: a

96. Soddy Fajan displacement laws govern

a) fusion

b) alpha beta decay

c) fission

d) scattering

Correct answer: b

97. Range velocity alpha beta gamma vacuum

a) $\alpha > \beta > \gamma$

b) $\beta > \gamma > \alpha$

c) $\gamma > \beta > \alpha$

d) $\alpha = \beta = \gamma$

Correct answer: c

98. Ionizing power order

a) $\gamma > \beta > \alpha$

b) $\beta > \gamma > \alpha$

c) $\alpha > \beta > \gamma$

d) $\gamma > \alpha > \beta$

Correct answer: c

99. Nuclear reactor steam drives

a) pump

b) turbine generator

c) compressor

d) fan

Correct answer: b

100. India's first atomic commission chairman

a) Vikram Sarabhai

b) Homi Bhabha

c) Raja Ramanna

d) Anil Kakodkar

Correct answer: b